Document Control

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1. Purpose

1.1. The purpose of this document is to detail the process for surfactant administration for neonates at North Devon District Hospital.

1.2. The policy applies to all medical staff involved in the care of neonates requiring surfactant administration.

1.3. Implementation of this policy will ensure that:

- Neonates requiring surfactant are identified in a timely manner
- Medical staff are aware of the correct procedure for prescribing and administering surfactant to neonates
- Problems associated with surfactant administration are avoided

2. Definitions

2.1. BNFc – British National Formulary for Children

2.2. INSURE – Intubate, Surfactant, Extubate

2.3. LISA – Less Invasive Surfactant Administration

2.4. MIST – Minimally Invasive Surfactant Administration

2.5. NDHT – North Devon Healthcare Trust

2.6. RDS – Respiratory Distress Syndrome

2.7. SCU – Special Care Unit

2.8. Curosurf® SPC - Summary of Product Characteristics

3. Responsibilities

Role of the Neonatal Nurse

3.1. The Neonatal Nurse is responsible for:

- Ensuring that they are aware of this guideline and the procedure surrounding the administration of surfactant
- Ensuring that they are competent in assisting the medical staff to administer surfactant
- Ensuring that they can assist in preparing the appropriate equipment required for surfactant administration

Role of Medical Staff

3.2. The Medical staff are responsible for:

- Ensuring that they are aware of this guideline and the procedure surrounding the administration of surfactant
- Ensuring that they are competent in the administration of surfactant
- Ensuring that they follow this guideline to ensure the safety of the neonate

4. Background

4.1. Respiratory distress syndrome (RDS) due to surfactant deficiency is the most common cause of respiratory failure in preterm infants (Sardesai et al, 2017).

4.2. Surfactant administration corrects this deficiency in lung surfactant, lowering the surface tension at the alveolar air/liquid interface preventing collapse during expiration. Surfactant administration improves ventilator requirements, air leak and survival rates in newborn infants at risk of respiratory distress syndrome (RDS). The benefits are maximised by giving the surfactant as early as possible

5. Curosurf® Dosage Calculation

5.1. Indications

- The natural surfactant used at North Devon Healthcare NHS Trust (NDHT) is Curosurf®, single dose vials of 240mg/3ml & 120mg/1.5ml.

  NOTE: This is a natural porcine surfactant. If parents are Jewish or Muslim please discuss use and gain informed consent. Some parents accept it as it is not 'per se' ingested

Curosurf® Indications are:

- *For the treatment, including early rescue, of Respiratory Distress Syndrome (RDS) or hyaline membrane disease in newborn babies*
- *Prophylactic use in premature infants requiring intubation for stabilisation at risk from RDS or with evidence of surfactant deficiency.*

- Surfactant must be administered to all infants less than 28 weeks gestation requiring intubation. This should occur prior to inflation breaths if possible.
- Surfactant administration should be considered in the following situations:
  - Infants of 28 to 29+6 weeks gestation requiring intubation on delivery suite or if the mother has not received antenatal corticosteroids
  - Infants who require intubation for RDS following admission to SCBU

5.2. Dosage

'Rescue' Dosage (treatment of a neonate (birth weight >700g with RDS) • 100–200 mg/kg, then 100 mg/kg every 12 hours if required, maximum 300–400 mg/kg per course.
Prophylactic dosage (prophylaxis of RDS in neonates of 24-31 weeks gestation) • 100–200 mg/kg, administer soon after birth, preferably within 15 minutes, then 100 mg/kg after 6–12 hours if required, then 100 mg/kg after 12 hours if required, and if neonate still intubated. Max 300–400 mg/kg per course.

(All doses per BNFc 2021 and Curosurf® SPC (medicines.org.uk, 2021)

(*The full dose should be given – but clinician discretion may be used with regards to using the nearest 'full vial' if applicable.)

6. Administration

INSURE is the recommended technique for the administration of surfactant within NDHT, and is linked to reduced rates of barotrauma, airway inflammation secondary to mechanical ventilation and the need for continued mechanical ventilation in late preterm infants (Sweet et.al. 2019). Dose should be administered by / under the supervision of a doctor via the **endotracheal** tube (ETT) as soon as possible after intubation. Consideration should be given to continuing mechanical ventilation in neonates requiring transfer to another unit.

Other methods of administration may be considered as appropriate; LISA/ MIST techniques using a thin catheter to administer surfactant slowly (over 3minutes) in patients who are breathing spontaneously and receiving non-invasive respiratory support (Sardesai et al, 2017). These methods are not currently used at NDHT and should only be undertaken following consultation with the tertiary neonatal centre.

Caffeine must be administered to the patient if using these methods.

Procedure

Preparation: • Inform parents and obtain informed consent • Consultant should be present if at all possible. • Prescribe correct Curosurf® dose (as above Table) • Prepare ventilator (if not already ventilated) and set to PC-AC with Volume Guarantee • Pre-oxygenate baby (if deemed appropriate) • Suction the baby (if needed) • Consider INSURE or LISA, discuss options with tertiary centre • Consider caffeine (mandatory for LISA and INSURE)
Monitoring: • Ensure adequate monitoring of O_2 saturations, heart rate and blood gases (plus etC02 if ventilated) throughout
Administration: • Warm vial in hands (if possible) and invert vial gently before drawing up prescribed dose • Use vial-syringe connector from surfactant administration pack to screw into rubber on top if vial, fit syringe and draw up desired volume • Intubate airway and confirm correct position using yellow colour change on pedi-cap. Ensure symmetrical chest rise, stable heart rate >100, sats >90%

• Measure the NG tube/catheter in the surfactant administration pack, and cut just short of the length of the ET tube • Double check position of the ET tube • Hold baby's head in the midline neutral position • Attach syringe of surfactant to the NG tube/catheter from administration pack • Insert the NG tube/catheter as far as possible into ET tube and administer all of the surfactant as rapidly as tolerated • Remember dead space in the NG tube/catheter and inject a small amount of air to clear any remaining Curosurf®.
After administration: • After administration hand ventilate or increase ventilation until the infant is stable • Once stable connect to ventilator or extubate in case of INSURE • Chest expansion may improve rapidly after the dose. Ventilator requirements may rapidly improve and ventilation should be adjusted promptly in response to changing requirements • Avoid suction for one hour following administration unless signs of airway obstruction occur • Check blood gas • Update parents and tertiary centre • Document Curosurf® use for Pharmacy to reorder and claim back costing

7. Side Effects

- Destauration/ bradycardia shortly after administration – usually resolves with IPPV
- Increased risk of pulmonary haemorrhage
- Obstruction of ETT by mucous secretions or surfactant have been reported
- Transient depression in cerebro-electrical activity lasting 2 – 10 minutes after administration of dose has been observed

8. Storage

- Curosurf® is stored in Delivery Suite and Special Care Unit fridges.
- An unopened, unused vial can be returned to the fridge if it has been exposed to temperatures of up to 25°C for a maximum of 24 hours. There may be only one such episode per vial. I**f either 24 hours or 25°C has been exceeded the vial should be disposed of.** Inform pharmacy if this situation occurs.
- Neonatal nurses check stock level and expiry dates monthly supervised by the Paediatric Pharmacist.

9. Monitoring Compliance with and the Effectiveness of the Guideline

Standards/ Key Performance Indicators

9.1. Key performance indicators comprise:

- Nice Neonatal Quality Standards
- NHS Toolkit for High Quality Neonatal Services
- National Neonatal Audit Programme
- NHS Standard Contract for Neonatal Critical Care

Process for Implementation and Monitoring Compliance and Effectiveness

9.2. Staff are informed of revised documentation. There is an expectation that staff are responsible to keep updated on any improvements to practice and deliver care accordingly.

9.3. Data is collected by use of Badger data base and can be used to generate output for clinical and operational benchmarking.

9.4. Intensive care incidents are monitored by the neonatal governance team and neonatal network. Incidents are reported by the Datix system and South West Neonatal Network incident reporting process.

9.5. Non-adherence is reviewed and action plans made if required. Discussion and reviews occur at Directorate meetings, Governance meetings and Ward meetings. Learning and action plans are cascaded at these meetings and improvements implemented. Key findings and learning points will be disseminated to relevant staff.

10. References

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- Plymouth Hospitals NHS Trust. Neonatal Intensive Care Guidelines. Surfactant Administration (2011).

- Sardesai, S., Biniwale, M., Wertheimer, F., Garingo, A. and Ramanathan, R. (2017). Evolution of surfactant therapy for respiratory distress syndrome: past, present and future. Pediatric research, 81 (1) 240-248
- Sweet, D.G., et. Al. (2019). European Consensus Guidelines on the Management of Respiratory Distress Syndrome. [Online] Available at: https://www.karger.com/Article/Fulltext/499361 [Accessed 19/9/21]

11. Associated Documentation

- Standard Operating Procedure for suctioning neonatal patients on
- Elective Endotracheal Intubation Guidelines for Neonates
- Neonatal Ventilation Guidelines